

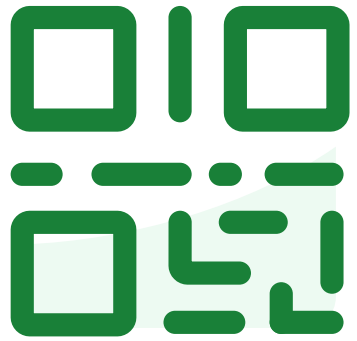
Lecture 4

K-Means and Probability

An introduction to unsupervised learning
and a review of core concepts in probability

EECS 189/289, Fall 2025 @ UC Berkeley
Joseph E. Gonzalez and Narges Norouzi

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Roadmap

- K-means Clustering
 - Scikit-Learn
 - Lloyd's Algorithm
 - Pixel K-Means
- Probability (Review)
 - Joint Distributions
- Wake Word Example

Questions



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K-means Clustering

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Prof. Gonzalez's Messy Biking Records



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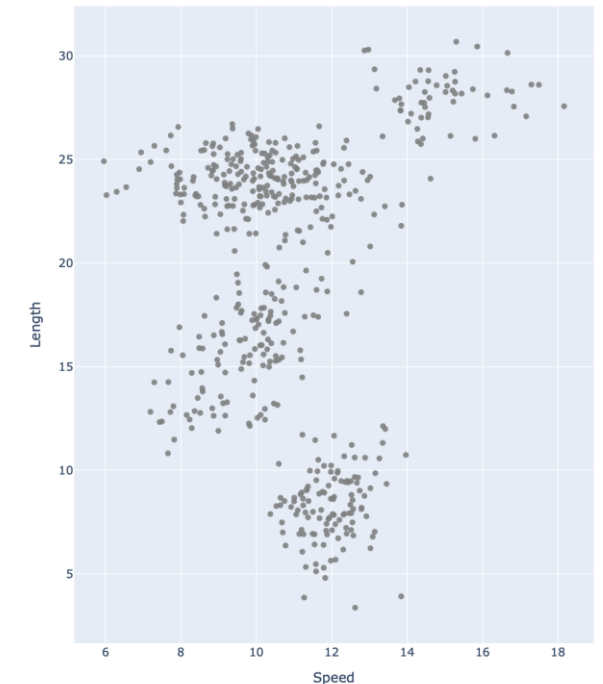
Professor Gonzalez has too many bikes



He has been recording the **speed** and **length** of his bike rides but not which bike he used.

He would like to answer questions like:

1. What is the **average ride time** for each bike?
2. Are there any rides that are **abnormally short or long**?
3. What is the **most likely bike** for each ride?



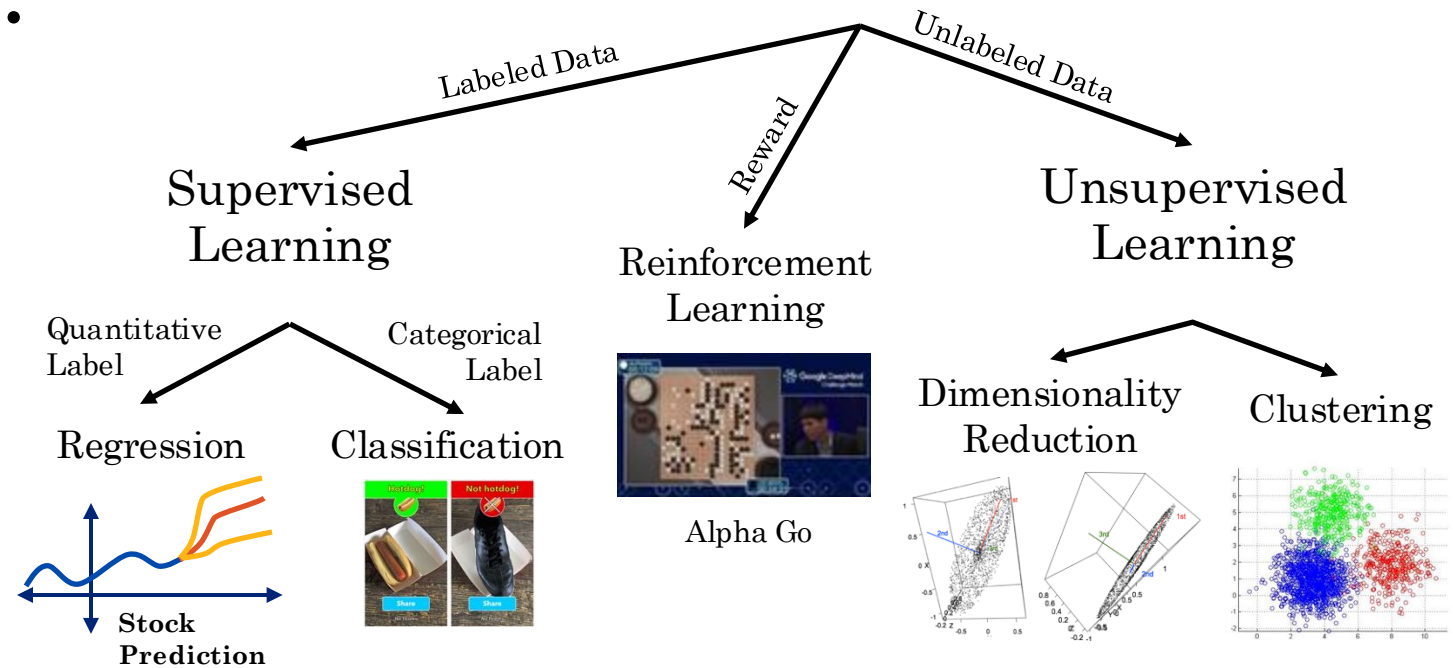
Learning Problem



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We have **unlabeled data** and we would like to divide the records into 4 **groups (clusters)** corresponding to the four bikes.

- **Unsupervised Learning:**
we don't have the labels.
- **Clustering:**
we are trying to infer the **unobserved (*latent*)** bike choice.



Let's try
k-means clustering

Demo

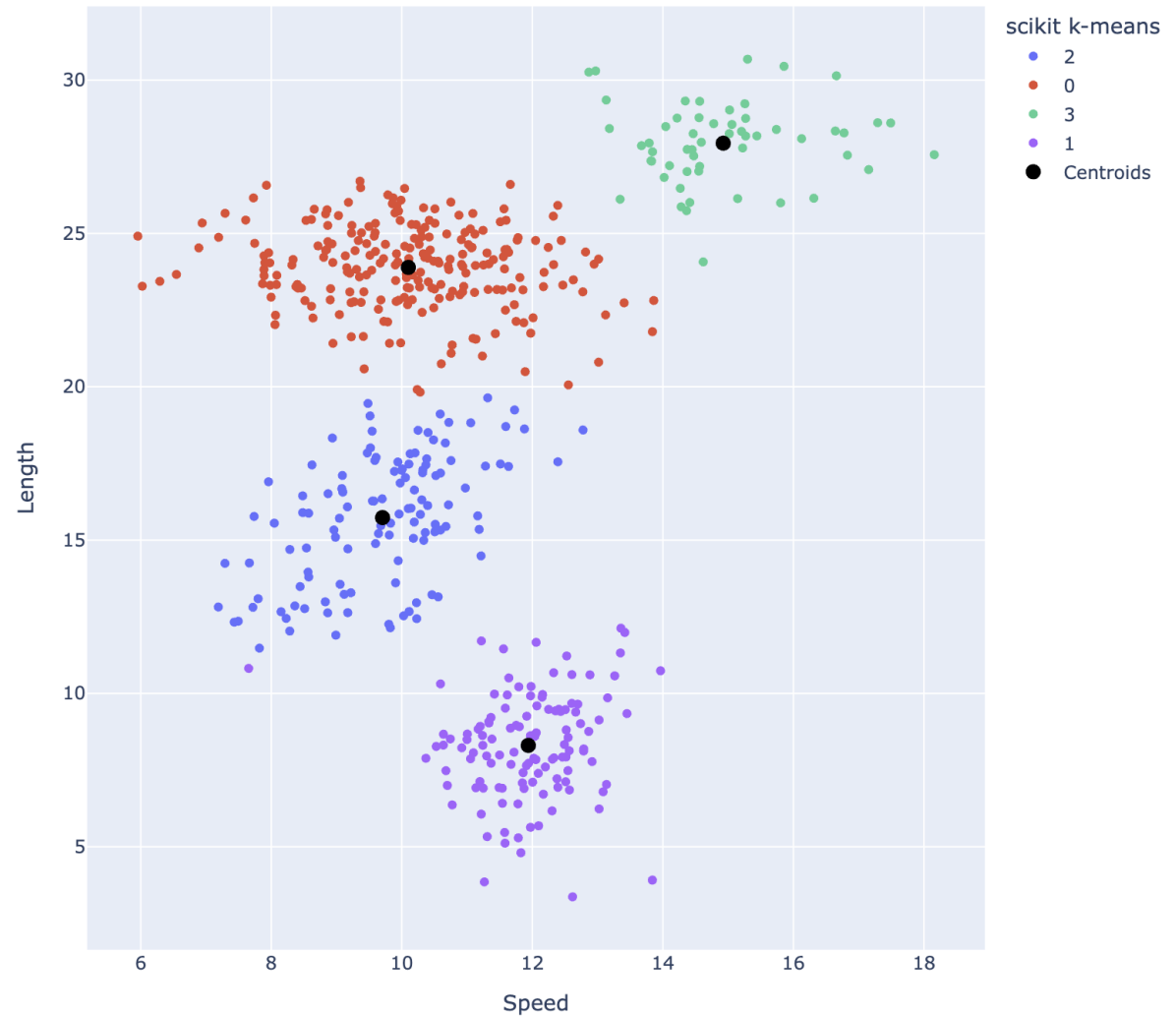
Using sklearn k-means clustering

SkLearn K-means Clustering



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K-Means Clustering of Bike Segments



Clustering With K-Means in Scikit-Learn



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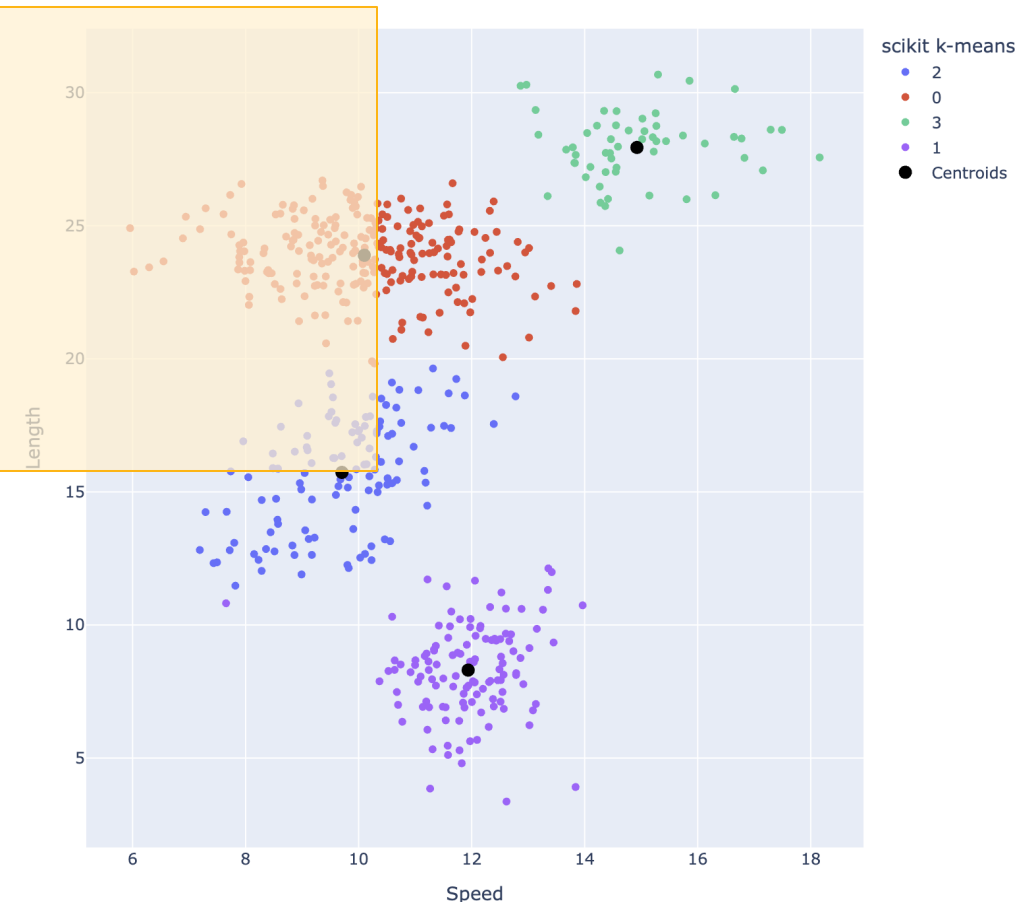
In the demo we wrote a few lines of code and obtained a **reasonable clustering of the ride-times**.

```
from sklearn.cluster import KMeans
# Create a KMeans model with 4 clusters
kmeans = KMeans(n_clusters=4, random_state=42)
# Fit the model to the data
kmeans.fit(bikes[['Speed', 'Length']])
# Predict the cluster assignments
bikes['c'] = kmeans.predict(bikes[['Speed', 'Length']])
```

Today we will learn how this algorithm works.

We will review concepts in probability and explore more general density estimation techniques.

K-Means Clustering of Bike Segments





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Lloyd's Algorithm

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K-Means Clustering

Input: A collection of data points $\mathcal{D} = \{x_1, \dots, x_N\}$ where $x_n \in \mathbb{R}^D$.

Output: K cluster centers $\mu_k \in \mathbb{R}^D$ and an **assignment** $z_n \in \{1, \dots, K\}$ of each data point to one of the cluster centers.

- The means μ_k are **model parameters** of the model (z_n can be computed from μ) that are **fit to the data**.
- K is a **hyperparameter** which we need to select.

Objective: Each data point should be close to its assigned center.

$$\arg \min_{\mu, z} \sum_{n=1}^N \|x_n - \mu_{z_n}\|_2^2$$

(L₂-Norm)²
Euclidean Distance Squared

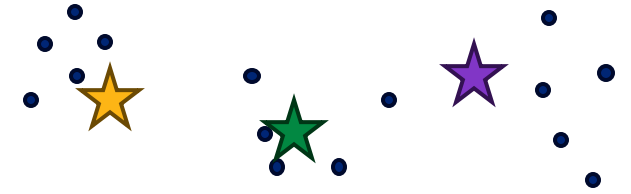
$$\|a\|_2^2 = \sum_{d=1}^D a_d^2$$



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K-Means Cluster (Lloyd's) Algorithm

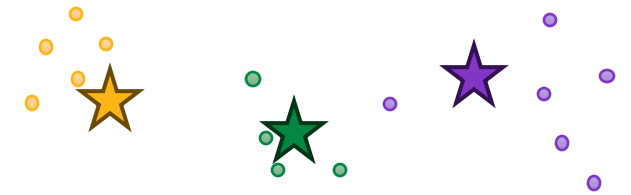
Initialization: Choose K points at random to be the initial cluster centers μ .



Iterate until convergence:

1. **Update Assignments:** Assign each point to its nearest cluster center.

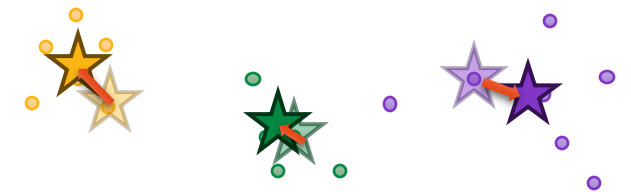
$$z = \arg \min_z \sum_{n=1}^N \|x_n - \mu_{z_n}\|_2^2$$



2. **Update Centers:** Recompute centers by averaging assigned points.

Why is this the mean of each cluster?

$$\mu = \arg \min_{\mu} \sum_{n=1}^N \|x_n - \mu_{z_n}\|_2^2$$





Updating the Cluster Centers

What is the value that minimizes?

$$\arg \min_{\mu} \sum_{n=1}^N \|x_n - \mu_{z_n}\|_2^2$$

We can re-arrange the objective to optimize with respect to μ :

$$\sum_{n=1}^N \|x_n - \mu_{z_n}\|_2^2 = \sum_{n=1}^N \sum_{d=1}^D (x_{nd} - \mu_{z_nd})^2 = \sum_{k=1}^K \sum_{i=1}^{N_k} \sum_{d=1}^D (x_{k_id} - \mu_{kd})^2$$

k_i is the i^{th} point
in cluster k

Sum over **points in cluster k**

Sum over **clusters**



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Minimizing the Transformed Objective

We take the derivative with respect to μ_{kd} :

$$\frac{\partial}{\partial \mu_{kd}} \sum_{k=1}^K \sum_{i=1}^{N_k} \sum_{d=1}^D (x_{k_id} - \mu_{kd})^2 = \sum_{i=1}^{N_k} -2(x_{k_id} - \mu_{kd})$$

Setting the derivative equal to zero and solving for μ_{kd} :

$$\sum_{i=1}^{N_k} -2(x_{k_id} - \mu_{kd}) = -2 \sum_{i=1}^{N_k} x_{k_id} + 2 N_k \mu_{kd} = 0$$

Cluster Mean
(Average Point)

$$\mu_{kd} = \frac{1}{N_k} \sum_{i=1}^{N_k} x_{k_id}$$



Is Lloyd's algorithm guaranteed to converge? Does it always produce an optimal clustering?



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Convergence of K-Means

Is the k-means (Lloyd's) algorithm guaranteed to converge?

- Yes 😊! Why?

Alternating minimization always decreases the objective

$$z = \arg \min_z \sum_{n=1}^N \|x_n - \mu_{z_n}\|_2^2 \quad \text{and} \quad \mu = \arg \min_{\mu} \sum_{n=1}^N \|x_n - \mu_{z_n}\|_2^2$$

How do we know when it has converged?

- The **cluster assignments** (z) stop changing.

Do the final z^* and μ^* minimize the objective?

- No ☹! Can be **local minima**.

Demo

Animated K-Means
Clustering

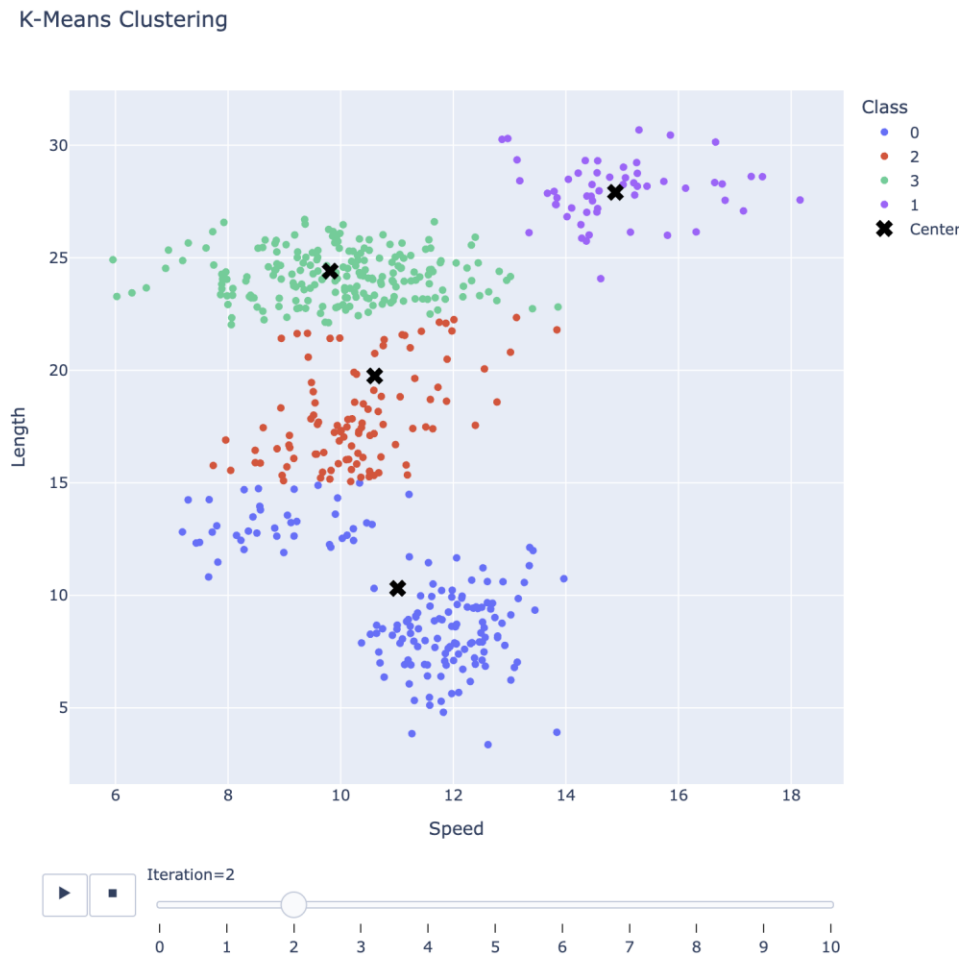
&

K-means for pixels

See live animation in the demo



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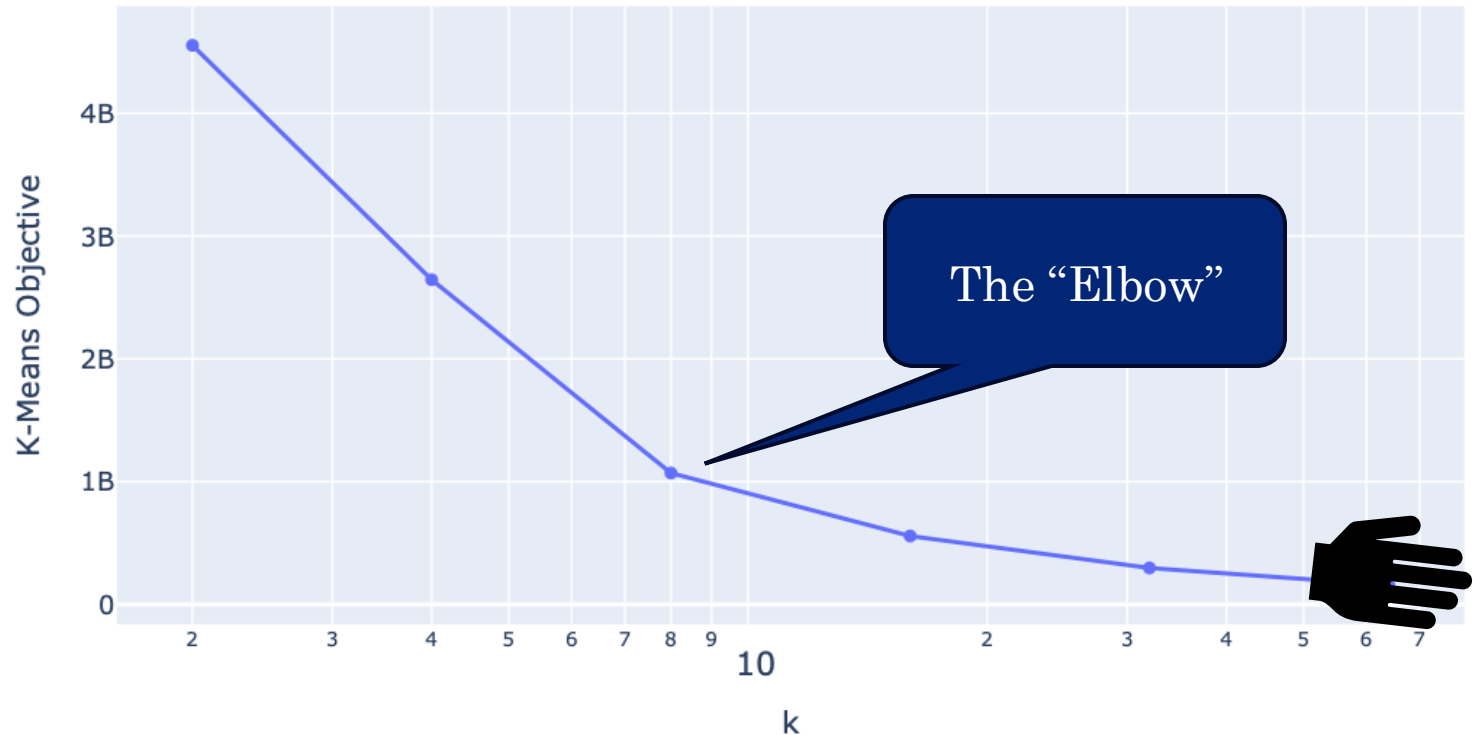
Choosing the Number of Clusters

There are several ways to select the **hyperparameter** K .

- Use **domain understanding** (e.g., 4 bikes)
- Use the “**elbow method**” to select the value of K with diminishing improvements in the objective:

$$\sum_{n=1}^N \|x_n - \mu_{z_n}\|_2^2$$

K-Means Objective vs Number of Clusters



Interpreting the Clusters

We used k-means to compute a **cluster assignment** for each ride.

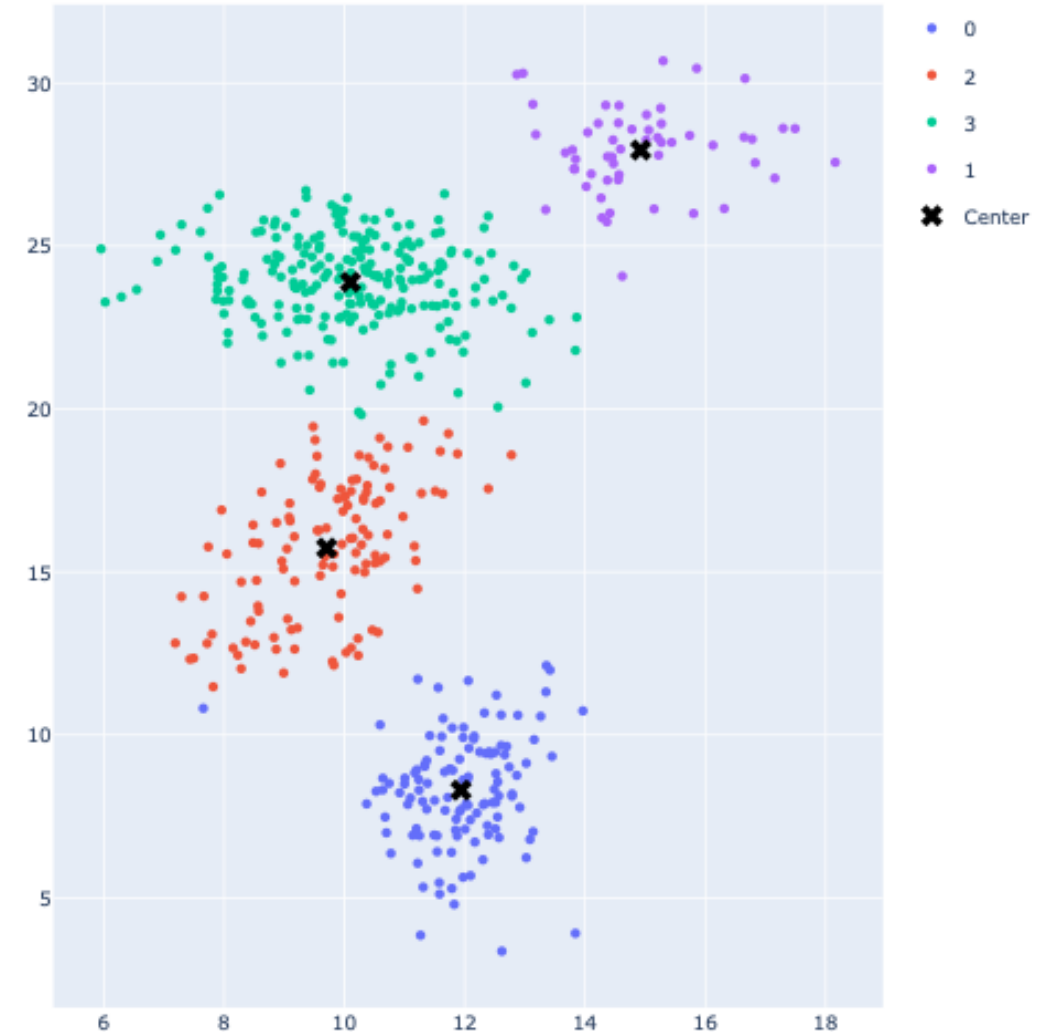
Which bike is each cluster?

- We don't know. Could ask for a few labels.

Does each cluster **represent a bike**?

- Maybe? But it could also correspond to **other factors** (e.g., group ride, traffic).

Use caution when interpreting clusters!





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Pixel K-Means

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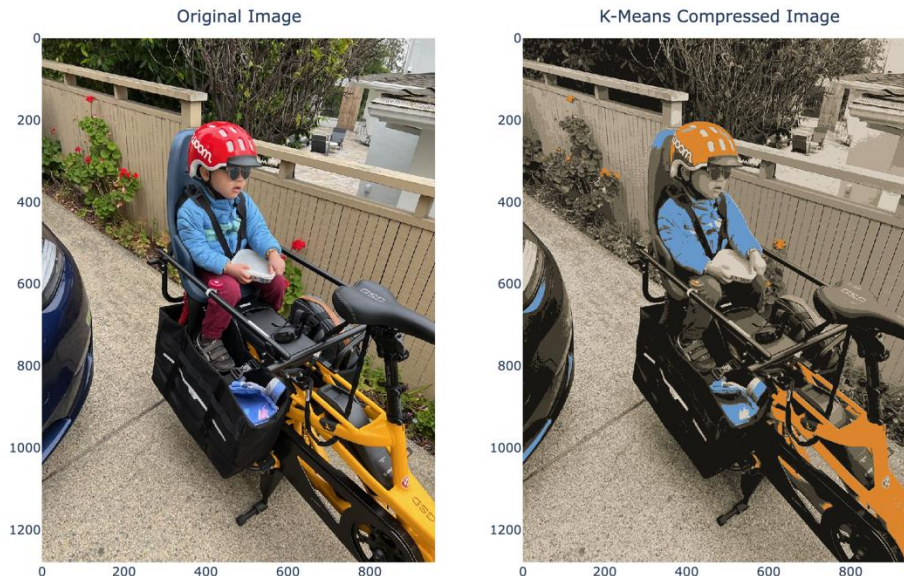
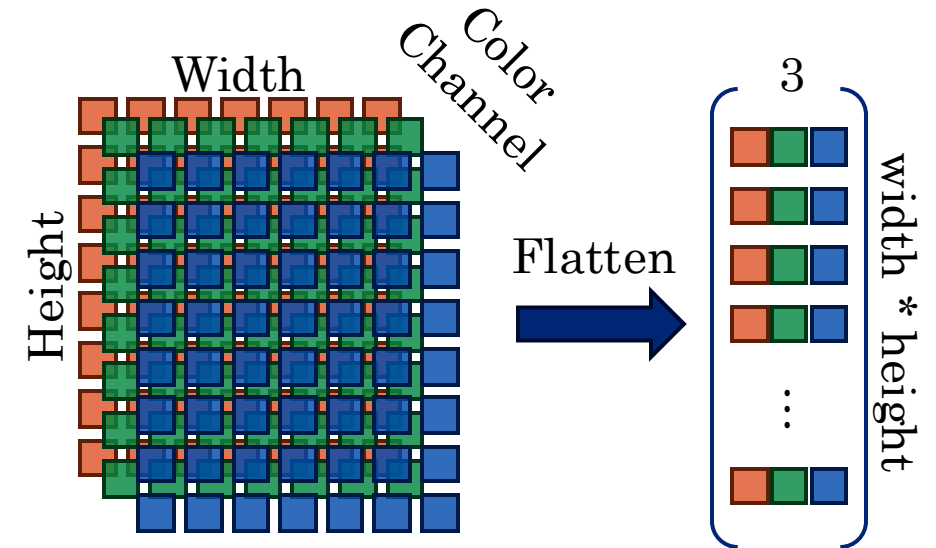
Pixel K-Means



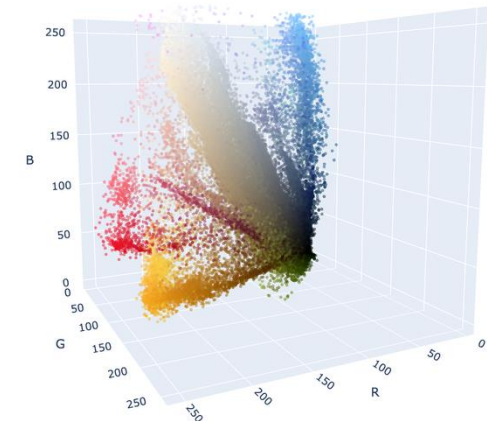
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The pixels in an image can be treated of as **vectors** in an **RGB vector space**.

- We can use k-means to compute the **clusters of colors** and render an image using just a few colors.



Pixels plotted in
RGB vector space.



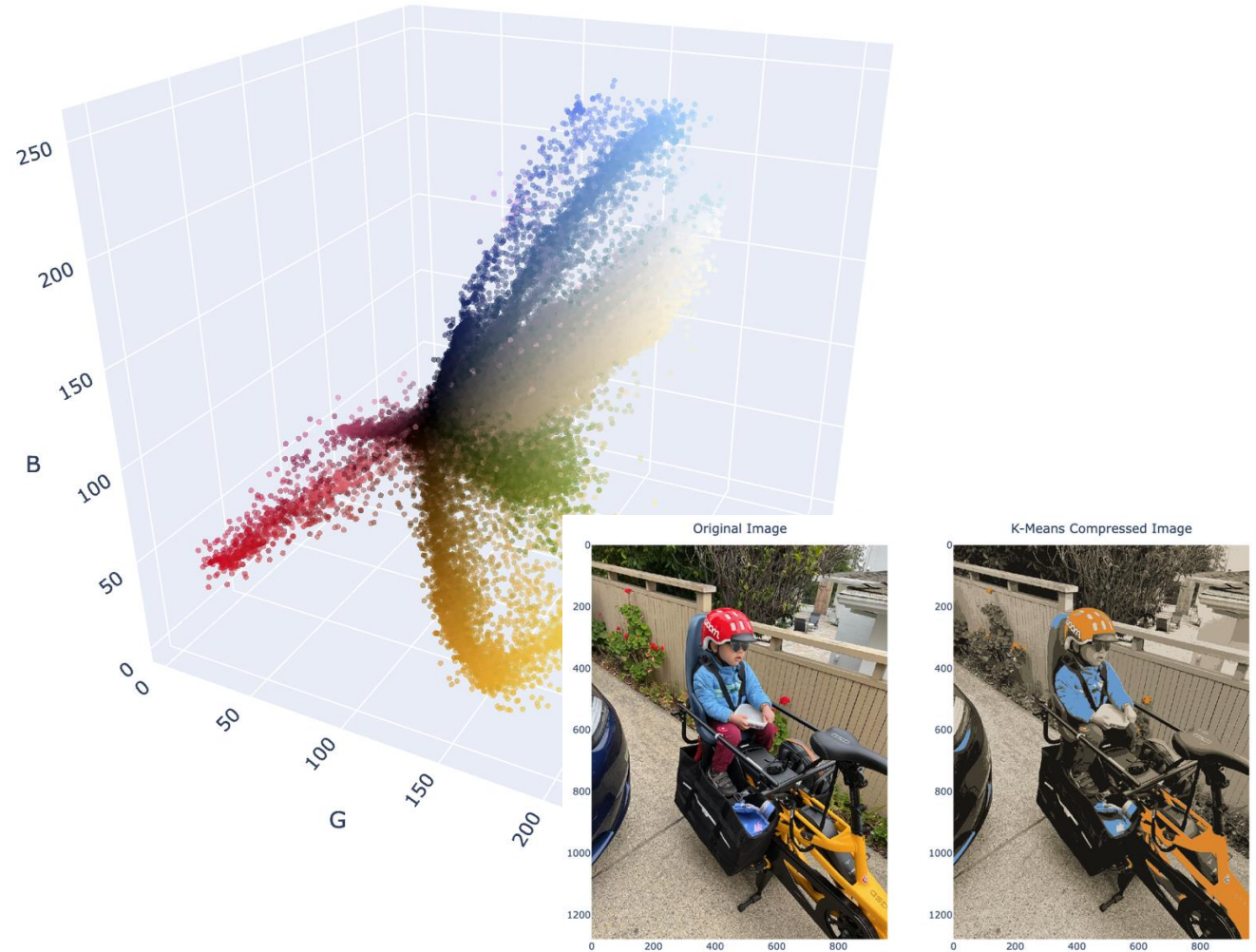


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Visualize Pixel Clustering

Demo

K-means on Pixels



Hard Cluster Assignments

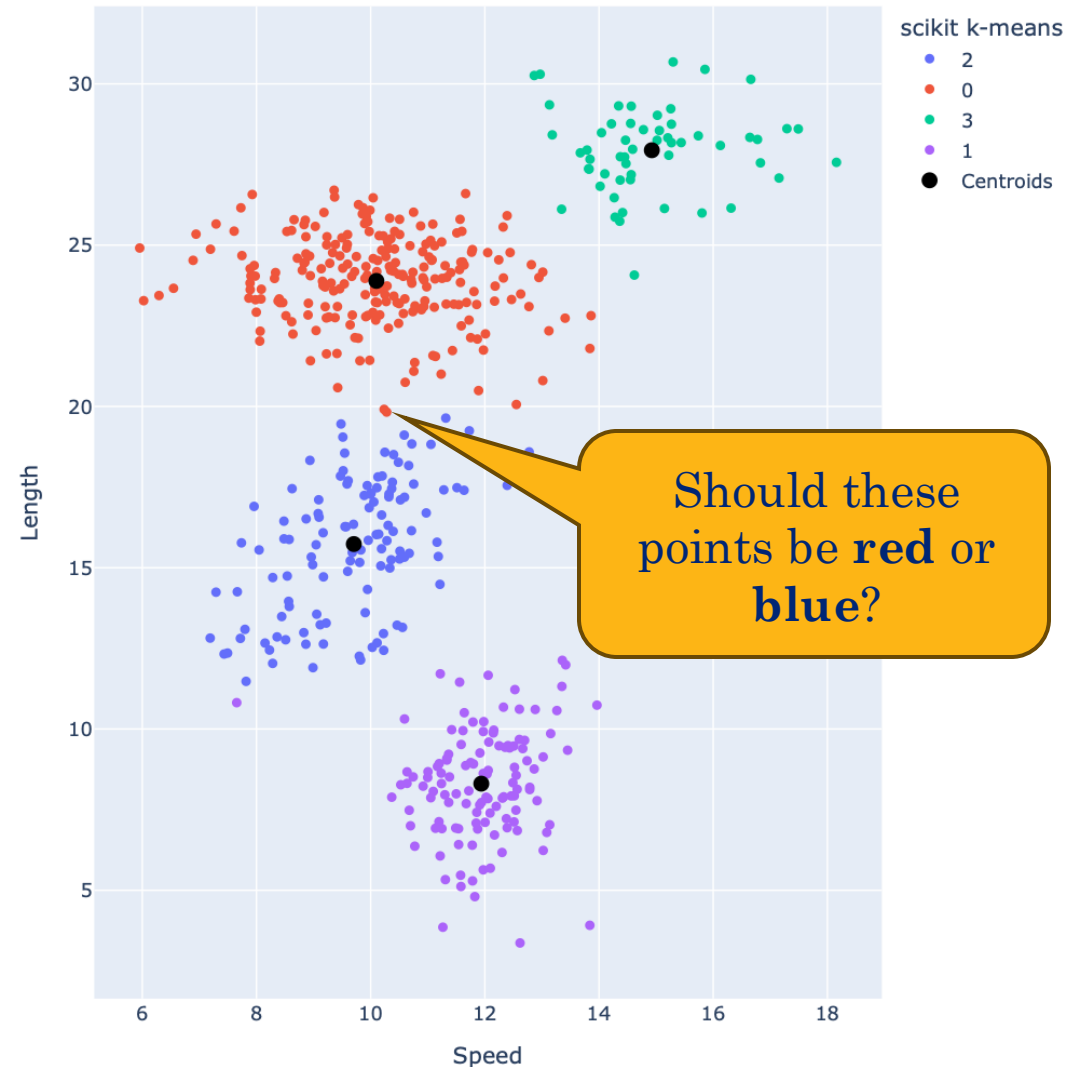


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K-means assigns each data point to **exactly one cluster**.

Do all the points belong in exactly one cluster?

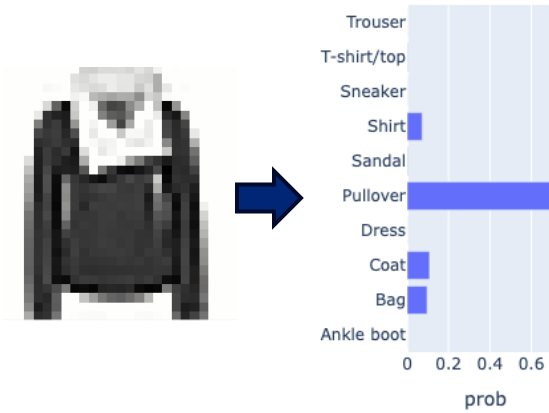
- **Maybe?** Each point represents one bike ride and therefore one bike.
- How could we at least measure the **uncertainty** in the **predictions**.



Uncertainty

ML is ultimately about **inference** – *making predictions*.

➤ Predictions are inherently *uncertain*.



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Source of Uncertainty:

- **Epistemic Uncertainty** (Reducible) – the *systematic uncertainty* that arises from a **finite training dataset** and our **modeling process**.
- **Aleatoric Uncertainty** (Irreducible) – the uncertainty that arises from observational *noise* in our **training data**.

Need a framework for uncertainty – *Probability!*



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Probability (Review)

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Probability



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Probability provides a framework for quantifying and manipulating uncertainty.

The probability of an event is:

- **Frequentist View:** The long-run relative frequency of an event in identical repeated trials.
- **Bayesian View:** The degree of belief (or plausibility) assigned to the event given the available information.

Which one is correct?



How do you interpret probabilities?

A brief review of the

Basics of Probability



The Joint Probability Distribution

Let X and Y be **discrete random variables** that take values:

$$X \in \{x_1, \dots, x_L\} \quad \text{and} \quad Y \in \{y_1, \dots, y_M\}$$

The joint probability function $p(X = x_i, Y = y_j)$ is the probability of the event $X = x_i$ and $Y = y_j$.

The joint probability satisfies the **following two properties**:

$$\forall x_i, y_j: p(X = x_i, Y = y_j) \geq 0 \quad \text{(Non-negativity)}$$

$$\sum_{i=1}^L \sum_{j=1}^M p(X = x_i, Y = y_j) = 1 \quad \text{(Normalization)}$$

$p(X = x_i, Y = y_j)$	y_1	y_2	
x_1	0.2	0.1	0.3
x_2	0.1	0.1	0.2
x_3	0.15	0.35	0.5
	0.45	0.55	1



The Joint Probability Distribution

Let X and Y be **discrete random variables** that take values

$$X \in \{x_1, \dots, x_L\} \quad \text{and} \quad Y \in \{y_1, \dots, y_M\}$$

The joint probability function $p(X = x_i, Y = y_j)$ is the probability of the event $X = x_i$ and $Y = y_j$.

The Sum Rule (Marginalization):
defines the distribution over a subset of the random variables.

$p(X = x_i, Y = y_j)$	y_1	y_2	$p(X = x_i)$
x_1	0.2	0.1	0.3
x_2	0.1	0.1	0.2
x_3	0.15	0.35	0.5
$p(Y = y_i)$	0.45	0.55	1

$$p(X = x_i) = \sum_{j=1}^M p(X = x_i, Y = y_j)$$



Conditional Probability

We define the conditional probability as the chance of observing $Y = y_j$ given **we observed** $X = x_i$

$$p(Y = y_j | X = x_i) = \frac{p(X = x_i, Y = y_j)}{p(X = x_i)}$$

Example:

What is the $p(Y = y_1 | X = x_3)$?

$$\frac{0.15}{0.5} = \frac{0.3}{0.7} = p(Y = y_i | X = x_3)$$

$p(X = x_i, Y = y_j)$	y_1	y_2	$p(X = x_i)$
x_1	0.2	0.1	0.3
x_2	0.1	0.1	0.2
x_3	0.15	0.35	0.5
$p(Y = y_i)$	0.45	0.55	1



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Empirical Probability Distributions

Consider N trials $\{(X_1, Y_1), \dots, (X_N, Y_N)\}$ in which we sample (X, Y) .

Let n_{ij} be the number (count) of trials where $(X = x_i, Y = y_j)$, we can construct a discrete **empirical probability distribution**:

$$\hat{p}(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

Frequentist View: In the limit as $N \rightarrow \infty$, the empirical probability distribution converges to the true probability distribution:

$$p(X = x_i, Y = y_j) = \lim_{N \rightarrow \infty} \hat{p}(X = x_i, Y = y_j)$$



The Product Rule of Probability

Recall the definition of the conditional:

$$p(Y = y_j | X = x_i) = \frac{p(X = x_i, Y = y_j)}{p(X = x_i)}$$

Multiply both sides by $p(X = x_i)$ you obtain the **product rule**:

$$p(X = x_i, Y = y_j) = p(Y = y_j | X = x_i) p(X = x_i)$$

The product rule is like a **chain rule** of probability – it allows us to **factorize** joint probability distributions into conditionals:

$$p(X, Y, Z) = p(Z | X, Y) p(X|Y) p(Y)$$

- *Notation*: we often drop x_i , y_j , and z_k for convenience.



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Independent Random Variables

Two variables X and Y are independent if the joint probability factorizes:

$$X \perp Y \Rightarrow p(X, Y) = p(X) p(Y)$$

Using the definition of conditionals:

$$X \perp Y \Rightarrow p(Y|X) = \frac{p(Y, X)}{p(X)} = \frac{p(Y)p(X)}{p(X)} = p(Y)$$

Independent and Identically Distributed (IID) data (X_i, Y_i) is:

$$p(X_1, Y_1, \dots, X_N, Y_N) = \prod_{n=1}^N p(X_n, Y_n)$$

- We will often assume IID data (though it is often not true in practice).



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Wake Word Example

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Example: Wake Words

A **wake word** is a verbal cue that triggers voice assistants to start actively listening.

Example: *“Alexa, set an alarm for 6:00AM”*

Most voice assistants **continuously** run a wake word detector model on every sound they hear.



Example: Wake Words

A **wake word** is a verbal cue that triggers voice assistants to start actively listening.

Example: *“Alexa, set an alarm for 6:00AM”*

Most voice assistants **continuously** run a wake word detector model on every sound they hear.



Example: Wake Words



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Streamed to the cloud
for processing.

How good does the wake word detector model need to be?

- To answer this question, we can model the **joint probability** of the **wake word detector** on an audio segment.

Before we listen to the audio segment, there are two random variables:

1. Was a wake word said: $X \in \{0, 1\}$ (0: no, 1: yes)
2. Was a wake word detected: $Y \in \{0, 1\}$ (0: negative, 1: positive)



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Analyzing the Wake Word Detector

Let's assume wake words are rare occurring less than 0.01% of segments:

$$p(X = 1) = 0.0001 \quad \text{and} \quad p(X = 0) = 1 - p(X = 1) = 0.9999$$

Suppose the wake word model **detects 99% of the wake words**:

$$p(Y = 1 | X = 1) = 0.99$$

And **falsely detects (false positive)** a wake word just 0.1% of the time:

$$p(Y = 1 | X = 0) = 0.001$$

If the **detector is positive**, what is the chance that **it was a wake word**?

$$p(X = 1 | Y = 1) = \frac{p(Y = 1, X = 1)}{p(Y = 1)} = \frac{p(Y = 1 | X = 1) p(X = 1)}{p(Y = 1)}$$

Bayes' Theorem



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Bayes' Theorem

Named after Thomas Bayes who was an 18th century statistician, philosopher, and minister.

$$p(A|B) = \frac{p(B|A)p(A)}{p(B)} = \frac{p(B|A)p(A)}{\sum_A p(B|A)p(A)}$$

- Derived from product rule: $p(A, B) = p(A|B)p(B) = p(B|A)p(A)$



Useful in Bayesian statistics:

- $p(A)$ is the **prior belief** about A
- $p(B|A)$ is the **likelihood** of an observation B given those beliefs
- $p(A|B)$ is our **posterior belief** about A given our observation B



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Analyzing the Wake Word Detector

We assumed rare wake words: $p(X = 1) = 0.0001$

A **true positive rate (recall)** of 99%: $p(Y = 1 | X = 1) = 0.99$

And a **false positive rate** of 0.1%: $p(Y = 1 | X = 0) = 0.001$

If the **detector is positive**, what is the chance that **it was a wake word**?

$$p(X = 1 | Y = 1) = \frac{p(Y = 1 | X = 1) p(X = 1)}{p(Y = 1 | X = 1) p(X = 1) + p(Y = 1 | X = 0) p(X = 0)}$$
$$p(X = 1 | Y = 1) = \frac{0.99 * 0.0001}{0.99 * 0.0001 + 0.001 * 0.9999} = 0.09 = 9\%$$

Is that good (enough)?

How would we improve it?



How could we improve the Wake Word Detector?

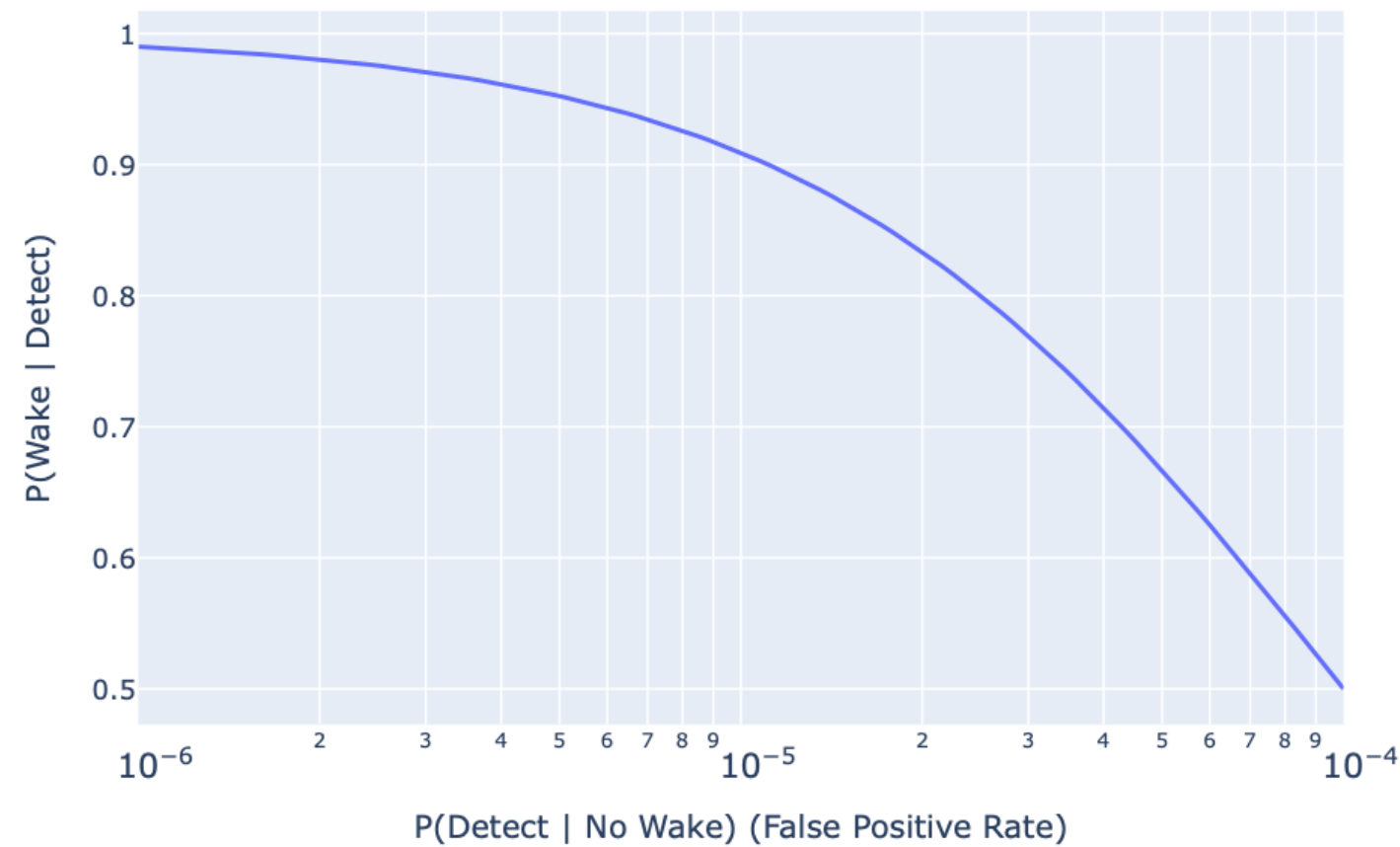


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Demo

Analysis of Recall and False Positive Rates

$P(\text{Wake} \mid \text{Detect})$ vs False Positive Rate for Perfect Recall





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Bayesian Updates: Wake Word Detector

Prior: we assumed rare wake words $p(X = 1) = 0.0001$

After observing the test $Y = 1$ we **updated our belief** about the presence of the wake word using Bayes Rule:

$$p(X = 1 | Y = 1) = \frac{p(Y = 1, X = 1)}{p(Y = 1)} = \frac{p(Y = 1 | X = 1) p(X = 1)}{p(Y = 1)} = 0.09$$

Posterior: the updated belief $p(X = 1 | Y = 1)$ after the observation.

Bayesian Updating: the process of starting from a **prior belief** and then using **observations** to compute **posterior distribution** (updated belief) is central to **Bayesian machine learning**.

Lecture 4

K-Means and Probability

Credit: Joseph E. Gonzalez and Narges Norouzi

Reference Book Chapters:

- **Clustering:** Chapter 15.1 (k-means)
- **Probability:** Chapter 2.[1-2]